HARSH GUJARATHI

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EDUCATION

Georgia Institute of Technology Atlanta, GA
Master of Science in Computer Science *August 2026 – 2028*

Birla Institute of Technology and Science, Pilani Goa, India
Bachelor of Engineering in Computer Science (Minor in Data Science); CGPA: 9.2/10.0 *November 2020 – June 2024*

RESEARCH EXPERIENCE

Undergraduate Research Assistant, Confluences of Computer Science Lab January 2023 — July 2024
Prof. Kunal Korgaonkar, BITS Pilani *Goa, India*

- Architected an **LLM-driven modernization framework** to parse complex, undocumented legacy codebases; parsing based on code specifications to understand and refactor legacy systems.
- Engineered a privacy-preserving middleware layer atop **Zulip**; deployed a **distributed testbed** to implement **mixnet-style message batching** and encrypted sender identifiers.
- Mitigated trace-through traffic analysis in controlled environments and identified critical **configuration enforcement vulnerabilities**, documenting the tension between cryptographic security and **message latency**.

Undergraduate Research Assistant, IoT Research Lab August 2021 — July 2024
Prof. Neena Goveas, BITS Pilani *Goa, India*

- Engineered a **reproducible Digital Twin** for campus solar installations, benchmarking gradient boosting models (XGBoost) against **Physics-Informed Neural Networks (PINNs)**.
- Achieved a **10% increase in predictive accuracy** with PINNs but identified a **5x penalty in model size**, quantifying the critical trade-off between model fidelity and **real-time deployment viability**.
- Modeled departmental TA allocation as an **NP-Hard Constraint Satisfaction Problem**; led a team of 7 to engineer heuristic scheduling algorithms that optimized assignments under strict course/student constraints.
- Developed *Divyang*, an Android app for prosthetic recommendations; deployed an end-to-end **MobileNet SSD** model finetuned via **TensorFlow Lite** to enable efficient inference on resource-constrained edge devices.

PROFESSIONAL EXPERIENCE

Software Engineer July 2024 — July 2026
Microsoft — Azure Migrate *Hyderabad, India*

- Delivered the **AI-First** experience for Azure Migrate Agent by integrating **RAG** with a self-built **NL2KQL** tool for custom KQL query generation, improving agent accuracy from **40% to 85%**.
- Owned the IISServer discovery and assessment microservice end-to-end — a distributed, workflow-driven service built on a **pub/sub model** with concurrent task orchestration; extended native coverage to **.NET Core, Python, PHP, and Java** apps across VMWare vCenter environments.
- Spearheaded the re-architecture of the compound assessment engine to integrate event-driven signals from **Azure Resource Graph**, enabling a more extensible and scalable assessment pipeline.
- Drove expansion of daily E2E test coverage across all **44 Azure regions**, safeguarding service reliability and data integrity for global enterprise-scale deployments.
- Sole owner of a critical fallback extension for non-container workloads; architected and shipped the feature, driving assessment coverage to **99.9%** across **100K+** enterprise workloads.
- On-call rotation across a Tier-1 global service; executed end-to-end RCAs and coordinated cross-team mitigations, reducing Sev-2 incidents and sustaining the microservice at a **99.9% SLA**.

Member of Technical Staff Intern January 2024 — June 2024
DevRev *Bengaluru, India*

- Strengthened the reliability of the Global Notifications microservice handling **60K+ daily events**, spanning rate limiting, throttling, and deduplication.
- Engineered distributed rate limiting via **Redis timed buckets** with event-based throttling, eliminating notification drops under bursty load.
- Built a fault-tolerant deduplication engine reducing spam by **85%**; implemented hierarchical clustering with $O(\log n)$ **lookup** via dendrogram traversal for topic-aware redundancy removal.

- Engineered an LLM-driven summarization pipeline (GPT-4 via LangChain) for digest generation, compressing data throughput by **6x** and improving actionable signal density.
- Drove performance profiling across high-throughput flows (reminders, tasks), surfacing bottlenecks and improving system reliability for global delivery.

Software Engineer Intern May 2023 — July 2023
Microsoft — Azure Backup and Site Recovery *Hyderabad, India*

- Instrumented customer-side meter telemetry for the Azure Business Continuity Center using **OpenTelemetry** and **Geneva**, enabling real-time observability for V1 services.
- Packaged the telemetry instrumentation as a reusable **NuGet library** with **99% test coverage**, adopted by adjacent engineering teams to accelerate integration for continuity scenarios.

SELECTED PROJECTS

IP-Wrapper October 2022
Developer *github.com/Neo-PL/IP-Wrapper*

- Developed a **multithreaded image processing engine** in C++ implementing custom convolution algorithms for operations like Gaussian blur, Sobel edge detection, and sharpening.
- Optimized performance using **OpenMP parallelization** to distribute channel processing and pixel computations, achieving efficient execution on multi-core architectures.
- Designed a modular **kernel-based architecture** allowing dynamic loading of custom filter masks and support for both monochromatic and RGB image streams.
- **Stack:** C++, OpenCV, OpenMP, CMake, Parallel Computing.

Amplitude-Phase Recombination (APR) April 2023
Deep Learning Developer *github.com/alphaNewrex/APR-Jupyter*

- Developed a novel **frequency-domain data augmentation** pipeline in **PyTorch** that synthesizes diverse training samples by recombining the amplitude and phase spectra of images via FFT/IFFT operations.
- Integrated the APR algorithm with state-of-the-art CNN architectures (**DenseNet, ResNet, WideResNet**) to enhance feature learning and model generalization on the CIFAR-10/100 datasets.
- Conducted extensive robustness benchmarking using the **CIFAR-10-C** dataset, demonstrating improved resilience against 15+ types of common image corruptions compared to standard augmentation baselines.
- **Stack:** Python, PyTorch, NumPy, Computer Vision, Fourier Transforms, Deep Learning.

TECHNICAL SKILLS

Languages	Python, C++, Java, C#, Go, SQL, TypeScript, JavaScript
ML & Deep Learning	PyTorch, TensorFlow, Scikit-learn, OpenCV, LangChain, RAG, LLM Agents, CNNs, FFT/IFFT
Distributed Systems	Kubernetes, Docker, Kafka, Pub/Sub, gRPC, Azure Resource Graph, OpenTelemetry
Data & Backend	PostgreSQL, MongoDB, Redis, .NET Core, Node.js, NumPy, Pandas

TEACHING EXPERIENCE

Teaching Assistant, Department of CSIS August 2022 — Dec 2023
BITS Pilani *Goa, India*

- **Courses:** Operating Systems, Computer Architecture (x2), Software for Embedded Systems, Data Structures & Algorithms, Object Oriented Programming, Logic in Computer Science.
- Facilitated lab sessions and tutorials for **7 undergraduate and graduate courses**, mentoring students on complex topics including **Linux system calls, Verilog/FPGA design, and MIPS architecture**.
- Designed comprehensive **lab questions and reference materials** for core systems courses, and evaluated assignments for graduate-level VLSI architectures.

Instructor, Centre for Technical Education March 2022 — June 2022
BITS Pilani *Goa, India*

- Designed the curriculum and taught a comprehensive **Android Development** course to a batch of 45 students, covering application lifecycle, UI/UX design, and backend integration.